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Roll No

BE-301 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Mathematics - II
(Common for all Branches)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions out of eight questions.
 ii) All questions carry equal marks.

1. Find the Fourier cosine transform of $f(x) = e^{-3x} + e^{-4x}$.

2. Find the Fourier series expansion of $f(x) = |x|$ in $(-\pi, \pi)$.

3. Using convolution theorem, find $L^{-1} \left\langle \frac{S}{(S^2 + a^2)^2} \right\rangle$.

4. Evaluate the following:

i) $L\{e^{3t} \sin at\}$ ii) $L\{t \sin 2t\}$

5. Solve the partial differential equation $x^2 p + y^2 q = nxy$

6. Solve in series the differential equation.

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$$

7. Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ by the method of separation of variables.

OR

Define divergence of a vector point function and explain its meaning.

8. Solve by Lagrange's method.

a) $x^2 p + y^2 q = z^2$

b) $xzp + yzq = xy$

OR

Use Stoke's theorem to evaluate $\int_c \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ and c is rectangle bounded by $x = \pm a, y = 0$ and $y = b$.
